

UNIT 2 – GAUSS'S LAW

2.1 Fields of Continuous Charge Distributions**Answer key**

Placeholder worksheet for the site prototype. Free for classroom use.

1. A thin rod of length L carries total charge Q spread uniformly along it. Find the electric field at a point on the rod's axis a distance a from the nearer end.

Let $\lambda = Q/L$ and put the near end at $x = 0$ with the field point at $x = -a$. A slice $dq = \lambda dx$ at position x is a distance $a + x$ away, so $dE = \frac{k\lambda dx}{(a+x)^2}$. Integrating: $E = k\lambda \int_0^L \frac{dx}{(a+x)^2} = k\lambda \left[\frac{1}{a} - \frac{1}{a+L} \right] = \frac{kQ}{a(a+L)}$, pointing away from the rod.

2. A ring of radius R carries charge Q uniformly. Find the field on the axis a distance z from the centre, and the value of z where it is largest.

By symmetry only the axial component survives: $E_z = \frac{kQz}{(z^2 + R^2)^{3/2}}$. Setting $dE_z/dz = 0$ gives $z^2 + R^2 - 3z^2 = 0$, so the maximum is at $z = R/\sqrt{2}$.